

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Isopropyl acetate

**CAS No** 108-21-4

**Synonyms** 2-Acetoxyp propane; 2-Propyl Acetate.

**Product Code** **C18080**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 2  
Category 3

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

AUH066 - Repeated exposure may cause skin dryness or cracking

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P312 - Call a POISON CENTER or doctor if you feel unwell

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information****Section 3 - Composition and Information on Ingredients**

| Component         | CAS No   | Weight % |
|-------------------|----------|----------|
| Isopropyl acetate | 108-21-4 | >95      |

**Section 4 - First Aid Measures**

|                     |                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                      |
| <b>Ingestion</b>    | Do NOT induce vomiting. Get medical attention.                                                                  |
| <b>Skin Contact</b> | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                       |
| <b>Eye Contact</b>  | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention. |

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting     |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Water mist may be used to cool closed containers. Chemical foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Water may be ineffective. Do not use a solid water stream as it may scatter and spread fire.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Specific Hazards Arising from the Chemical

Flammable. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Take precautionary measures against static discharges. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Wash hands before breaks and immediately after handling the product.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component         | Australia                                                                                    | New Zealand WEL                                                                              | ACGIH TLV                     | The United Kingdom                                         | Germany                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Isopropyl acetate | STEL: 310 ppm<br>STEL: 1290 mg/m <sup>3</sup><br>TWA: 250 ppm<br>TWA: 1040 mg/m <sup>3</sup> | TWA: 250 ppm<br>TWA: 1040 mg/m <sup>3</sup><br>STEL: 310 ppm<br>STEL: 1290 mg/m <sup>3</sup> | TWA: 100 ppm<br>STEL: 150 ppm | STEL: 200 ppm 15 min<br>STEL: 849 mg/m <sup>3</sup> 15 min | TWA: 100 ppm (8 Stunden). MAK<br>TWA: 420 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 200 ppm<br>Höhepunkt: 840 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| PVA            | < 120 minutes     | 0.3 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Butyl rubber   | < 20 minutes      | 0.35 mm         |                 |                                                                                |
| Nitrile rubber | < 20 minutes      | 0.38 mm         |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger

of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter conforming to EN371 (or AUS/NZ equivalent)

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                    |                                             |
|------------------------------------------------|------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless                          |                                             |
| <b>Physical State</b>                          | Liquid                             |                                             |
| <b>Odor</b>                                    | vinegar-like                       |                                             |
| <b>Odor Threshold</b>                          | 0.5 - 42 ppm                       |                                             |
| <b>pH</b>                                      | No information available           |                                             |
| <b>Melting Point/Range</b>                     | -73 °C / -99.4 °F                  |                                             |
| <b>Softening Point</b>                         | No data available                  |                                             |
| <b>Boiling Point/Range</b>                     | 88.8 °C / 191.8 °F                 |                                             |
| <b>Flash Point</b>                             | 4 °C / 39.2 °F                     | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available                  |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                     | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.8<br><b>Upper</b> 8 |                                             |
| <b>Vapor Pressure</b>                          | 61 mbar @ 20 °C                    |                                             |
| <b>Vapor Density</b>                           | 3.5                                | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.872                              |                                             |
| <b>Bulk Density</b>                            | Not applicable                     | Liquid                                      |
| <b>Water Solubility</b>                        | 31 g/L (20°C)                      |                                             |
| <b>Solubility in other solvents</b>            | No information available           |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                    |                                             |
| <b>Component</b>                               | <b>log Pow</b>                     |                                             |
| Isopropyl acetate                              | 1.03                               |                                             |
| <b>Autoignition Temperature</b>                | 460 °C / 860 °F                    |                                             |
| <b>Decomposition Temperature</b>               | No data available                  |                                             |
| <b>Viscosity</b>                               | 0.49 cP at 25 °C                   |                                             |
| <b>Explosive Properties</b>                    |                                    | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available           |                                             |
| <b>Other information</b>                       |                                    |                                             |
| <b>Molecular Formula</b>                       | C5 H10 O2                          |                                             |
| <b>Molecular Weight</b>                        | 102.13                             |                                             |

## Section 10 - Stability and Reactivity

**Reactivity** None known, based on information available

**Stability** Stable under normal conditions. Moisture sensitive.

|                                         |                                                                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition, Incompatible products, Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Acids, Bases.                                                                                                            |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                                                 |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                 |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

**Product Information** No acute toxicity information is available for this product

**(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component         | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation                    |
|-------------------|---------------------------|-------------------------------|------------------------------------|
| Isopropyl acetate | LD50 = 3000 mg/kg ( Rat ) | LD50 > 17436 mg/kg ( Rabbit ) | 50600 mg/m <sup>3</sup> , 8h (Rat) |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Category 3

**Results / Target organs**

Central nervous system (CNS)

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

|                                 |                                                                                                                                                                                              |                             |                               |          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------|----------|
| Ecotoxicity effects             |                                                                                                                                                                                              | Do not empty into drains. . |                               |          |
| Component                       | Freshwater Fish                                                                                                                                                                              | Water Flea                  | Freshwater Algae              | Microtox |
| Isopropyl acetate               | 265 mg/l LC50 48h                                                                                                                                                                            |                             |                               |          |
| Persistence and Degradability   | Readily biodegradable                                                                                                                                                                        |                             |                               |          |
| Persistence                     | Persistence is unlikely, based on information available.                                                                                                                                     |                             |                               |          |
| Bioaccumulative Potential       | Bioaccumulation is unlikely                                                                                                                                                                  |                             |                               |          |
| Component                       | log Pow                                                                                                                                                                                      |                             | Bioconcentration factor (BCF) |          |
| Isopropyl acetate               | 1.03                                                                                                                                                                                         |                             | No data available             |          |
| Mobility                        | The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air |                             |                               |          |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors                                                                                                                    |                             |                               |          |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance                                                                                                                               |                             |                               |          |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance                                                                                                                               |                             |                               |          |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                             |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.                   |

## Section 14 - Transport Information

### IMDG/IMO

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN1220            |
| <b>Proper Shipping Name</b> | ISOPROPYL ACETATE |
| <b>Hazard Class</b>         | 3                 |
| <b>Packing Group</b>        | II                |

### ADG

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN1220            |
| <b>Proper Shipping Name</b> | ISOPROPYL ACETATE |
| <b>Hazard Class</b>         | 3                 |
| <b>Packing Group</b>        | II                |

| Component                             | Hazchem Code |
|---------------------------------------|--------------|
| Isopropyl acetate<br>108-21-4 ( >95 ) | 3YE          |

### IATA

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN1220            |
| <b>Proper Shipping Name</b> | ISOPROPYL ACETATE |
| <b>Hazard Class</b>         | 3                 |
| <b>Packing Group</b>        | II                |

|                        |                                 |
|------------------------|---------------------------------|
| Environmental hazards  | No hazards identified           |
| Special Precautions    | No special precautions required |
| Additional information | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                    | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|------------------------------|-------------------------------------------------------------|------------------------|
| Isopropyl acetate - 108-21-4 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

#### **National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**

| Component         | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCs | ISHL | IECSC | KECL     |
|-------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Isopropyl acetate | X    | X     | 203-561-1 | -      | X    | X   | -    | X     | X    | X    | X     | KE-21670 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### **International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance



Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their disposal  
Not applicable.

| Component         | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Isopropyl acetate | 108-21-4 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Isopropyl acetate | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

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Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.  
Chemical incident response training.

**Revision Date** 20-Nov-2022  
**Revision Summary** Initial Release.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**